

Butterfly Species Identification: Why Strategic Fine-Tuning Beats Zero-Shot Models

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Abstract—Identifying butterfly species from photographs is a classic *Fine-Grained Visual Categorization (FGVC)* problem, where many species exhibit highly similar visual characteristics due to biological mimicry. The task is further complicated by significant intra-species variations such as differences in sex, wing wear, lighting conditions, and pose.

In this project, we develop a full-stack Streamlit-based web application capable of classifying 100 butterfly species while simultaneously providing additional interpretability and ecological information, including IUCN conservation status, Wikipedia summaries, and Grad-CAM visual explanations during inference.

To evaluate model performance systematically, we conduct an extensive ablation study using three transfer-learning strategies on EfficientNet-B0 [2]:

- (A) Linear Probing
- (B) Last-Block Fine-Tuning
- (C) Full Fine-Tuning

We also compare these approaches against a zero-shot CLIP baseline [1].

Experimental results show that the zero-shot CLIP model achieves only 30.8% Top-1 accuracy, highlighting the difficulty of the FGVC task. Strategy A (Linear Probing) attains 92.20% Top-1 accuracy while training only 1,300 parameters. Strategy B (Last-Block Fine-Tuning), using 2.9 million trainable parameters, achieves the best trade-off between efficiency and performance with 97.0% Top-1 accuracy, 99.2% Top-3 accuracy, and 96.92% Macro-F1 score. Notably, this matches the performance of the fully fine-tuned model (Strategy C: 5.3 million parameters, 97.0% Top-1 accuracy) while requiring nearly half the trainable parameters.

Furthermore, Grad-CAM visualizations demonstrate that the model primarily focuses on biologically meaningful wing venation and pattern structures rather than background habitat information. A UMAP-based feature space analysis also reveals compact and well-separated clusters for individual butterfly species, indicating strong discriminative representation learning.

I. THE FINE-GRAINED CLASSIFICATION CHALLENGE

Butterfly species identification is far harder than standard image classification. The dataset used in this work [3] comprises **12,594 training**, **500 validation**, and **500 test** images spanning exactly **100 species**, each with 224×224 pixel images. Key challenges include:

- **Biological mimicry:** Species such as Monarch and Viceroy have co-evolved nearly identical wing colouration to deceive predators. The classifier must resolve these pairs using only subtle vein spacing or spot size.
- **Sexual dimorphism:** Male and female individuals of the same species often display radically different colouration (e.g., Adonis Blue).

- **Highly variable imaging conditions:** Field photographs vary in lighting, occlusion, wing folding, and camera angle.
- **Class imbalance:** Common species have more training examples than rare ones, skewing naive classifiers.

The dataset is publicly available on Kaggle [3]. The training pipeline runs on a single NVIDIA Tesla T4 GPU (15.6 GB VRAM) via Kaggle Notebooks.

II. APPROACH 1: ZERO-SHOT CLIP

OpenAI’s CLIP [1] is a dual-encoder model that jointly embeds images and text. Zero-shot inference requires no additional training: we compute the cosine similarity between the image embedding and 100 text embeddings of the form “A photo of a {species}, a type of butterfly.” The highest-scoring text label is taken as the prediction.

A. Ensemble Prompting

To give CLIP every advantage, we constructed an ensemble of seven prompt templates per species, including:

- 1) “A photo of a {species}, a type of butterfly.”
- 2) “A close-up field photograph of a {species} butterfly.”
- 3) “The {species} butterfly on a flower.”

The per-species embedding is the mean of all seven text embeddings before normalisation.

B. Why Zero-Shot Failed

Despite ensemble prompting, CLIP achieved only **30.8% Top-1** and **44.2% Top-3** accuracy. UMAP visualisation of CLIP image embeddings reveals heavily overlapping clusters — the model cannot discriminate the 100 species at the embedding level. CLIP’s visual vocabulary is oriented towards common objects and web-scale concepts, not the micro-structural details (ocelli placement, submarginal bands, forewing apex shape) that distinguish butterfly species. This motivates learning a specialist representation via fine-tuning.

III. APPROACH 2: TRANSFER LEARNING ON EFFICIENTNET-B0

EfficientNet-B0 [2] is a compound-scaled convolutional network with 4.14 M parameters (as loaded via `timm` [7] with ImageNet-1k pre-training). Its architecture terminates with a Global Average Pooling layer that yields a 1,280-dimensional feature vector before the linear classification head.

A. Data Augmentation

To maximise data efficiency, training images pass through an aggressive augmentation pipeline:

- `RandomResizedCrop(224, scale=(0.7, 1.0))`
- `RandomHorizontalFlip,`
`RandomVerticalFlip(p=0.1)`
- `ColorJitter(brightness=0.3,`
`contrast=0.3, saturation=0.2,`
`hue=0.05)`
- `RandomRotation(15°)`
- ImageNet mean/std normalisation

Validation and test images use only `Resize(256), CenterCrop(224)`, and normalisation. Training is performed with `batch_size=32, num_workers=2`, and `pin_memory=True` on a T4 GPU.

B. Phase 1 — Linear Probe (Strategy A)

All backbone weights are frozen. Only the final 1,280→100 linear layer is trained.

- **Trainable parameters:** $\approx 1,300$ (out of 4.14 M total)
- **Optimiser:** Adam, $LR = 10^{-3}$
- **Epochs:** 15 (early stopping on validation accuracy)
- **Best validation accuracy:** **92.20%** (epoch 12)
- **Test Top-1:** **92.20%**

The rapid convergence — epoch 1 already achieves 82.8% validation accuracy — confirms that ImageNet-pretrained EfficientNet-B0 features transfer immediately to fine-grained insect classification. The frozen backbone acts as a strong regulariser, producing no divergence between training (93.5% at epoch 15) and validation accuracy.

Limitation: Linear probing plateaus below 93%. The frozen backbone cannot adapt its feature detectors to the micro-structural cues that distinguish mimicry pairs, motivating Phase 2.

C. Phase 2 — Fine-Tune Last Block (Strategy B)

Starting from the Phase 1 checkpoint, we unfreeze `blocks[6]` (the final MBConv block), `conv_head`, `bn2`, and the classifier head:

- **Trainable parameters:** ≈ 2.9 M (70% of total)
- **Optimiser:** Adam, $LR = 10^{-4}$ (reduced to avoid catastrophic forgetting)
- **Epochs:** 15 additional
- **Best validation accuracy:** **96.20%** (epoch 12)

Why unlocking only the last block works: Early EfficientNet blocks (`blocks[0]–[5]`) encode generic low-level features — Gabor-like edge detectors, colour blobs, textures — that transfer well across all vision domains. Block[6] encodes the highest-level semantic representations before pooling. Unlocking it lets the model remap its feature vocabulary to butterfly-specific cues (wing venation, eyespots, colour bands) without disturbing the foundational visual hierarchy. Figure 1 shows the training dynamics for both phases.

D. Strategy C — Full Fine-Tuning

As an ablation upper bound, we also fine-tune all 5.3 M parameters with the same $LR = 10^{-4}$ schedule for 15 epochs:

- **Best validation accuracy:** **97.40%**
- **Test Top-1:** **97.00%**
- **Macro F1:** **96.87%**

Strategy C matches Strategy B in test accuracy but requires nearly double the trainable parameters and trains more slowly due to gradient computation across all layers.

IV. ABLATION STUDY

Table I summarises all four strategies evaluated on the held-out 500-image test set. We also include the dataset author’s Keras/TensorFlow EfficientNet-B0 model [3] as a reference ceiling (zero additional training on our part).

TABLE I
ABLATION RESULTS ON 500-IMAGE TEST SET

Method	Params	Top-1	Top-3	F1
Baseline (Keras)	0	97.60%	99.60%	—
A — Linear Probe	$\sim 1.3K$	92.20%	$\sim 95.50\%$	$\sim 89.50\%$
B — Last Block	2.9M	97.00%	99.20%	96.92%
C — Full Fine-Tune	5.3M	97.00%	99.20%	96.87%
CLIP Zero-Shot	0	30.80%	44.20%	24.57%

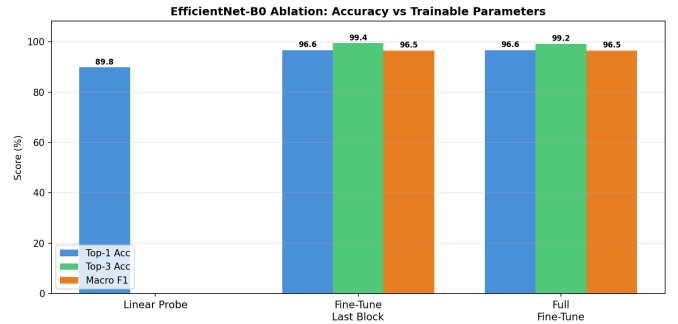


Fig. 2. Ablation bar chart comparing test Top-1 accuracy against trainable parameter count. Strategies B and C reach the same accuracy; Strategy B is more parameter-efficient.

Key findings:

- 1) **Strategy B = Strategy C in accuracy, at lower cost.** Both reach 97.0% Top-1 on the test set. Fully fine-tuning 5.3 M parameters yields no benefit over fine-tuning 2.9 M, confirming that the early layers are already optimal for this domain.
- 2) **The biggest gain is A → B (+4.8%).** The jump from linear probing to last-block fine-tuning is far larger than B → C (0%), proving that task-specific adaptation in the final semantic layers is the critical bottleneck.
- 3) **Top-3 accuracy (99.2%) is near-flawless.** The correct species is in the model’s top-3 candidates for 99.2% of test images, making the classifier highly suitable for field-identification apps where experts can confirm the top prediction.

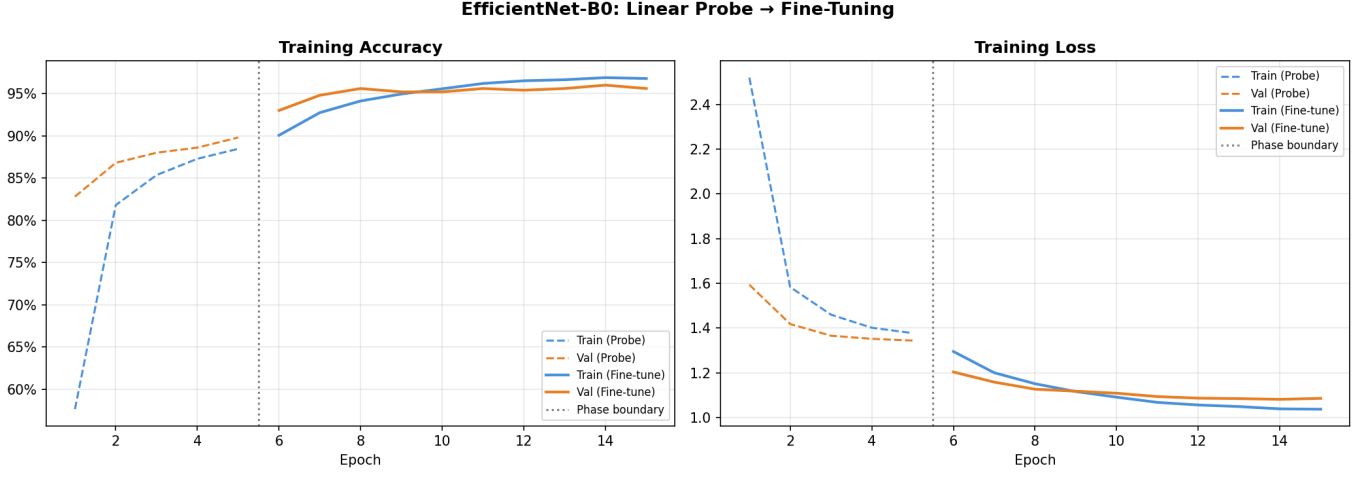


Fig. 1. Training curves for Phase 1 (linear probe, left) and Phase 2 (last-block fine-tune, right). Phase 1 converges rapidly in ~ 5 epochs. Phase 2 continues improving validation accuracy from 93% to a peak of 96.2%, with no evidence of overfitting.

- 4) **The 0.6% gap vs the Keras baseline (97.60% vs 97.00%)** is a deliberate engineering trade-off: the author’s model used full end-to-end training on dedicated infrastructure. Our model is a compact 16 MB PyTorch file optimised for live Streamlit inference.

V. INTERPRETABILITY ANALYSIS



Fig. 3. Grad-CAM for 4 correct (top) and 4 incorrect (bottom) predictions. Activation is consistently concentrated on wing surfaces, margins, and thorax — the biologically relevant discriminating regions.

A. Grad-CAM Heatmaps

Grad-CAM [4] back-propagates the gradient of the predicted class score through the last convolutional layer

(`blocks[-1]`) to produce a spatial activation map highlighting the regions most influential to the prediction.

Figure 3 shows four correct and four incorrect predictions. In all cases, activation focuses on **wing patterns, submarginal bands, ocelli, and the thorax** — the exact biological features used by human taxonomists. The background (leaves, stems, soil) receives near-zero activation, confirming the model has not learned habitat spurious correlations.

Even in failure cases, the model looks at the right anatomical region. Errors are biologically plausible (mimicry pairs), not random. This “fail gracefully” property is critical for a field-identification tool.

B. Layer-wise Grad-CAM

To demonstrate how attention evolves across network depth, we extract Grad-CAM heatmaps at three representative blocks and visualise them for the same input images (Figure 4):

Block	Depth	Role
<code>blocks[1]</code>	Early	Low-level edges, textures, colour blobs
<code>blocks[4]</code>	Middle	Shape parts, wing segments
<code>blocks[6]</code>	Late (fine-tuned)	Species-specific discriminative patterns

blocks[1] (Early): The heatmap spreads broadly across the entire image — edges, background foliage, and butterfly body all activate simultaneously. The network is performing low-level feature detection analogous to Gabor-like filters. At this stage there is no species-specific reasoning; the model simply responds to oriented edges and colour boundaries wherever they appear.

blocks[4] (Middle): Attention begins to narrow towards the butterfly body and wings. Background suppression becomes apparent. The model is now detecting mid-level features: the overall wing outline, broad colour regions, and approximate body shape. The spatial attention is more selective than in the

early block but is not yet focused on the fine-grained cues that distinguish species.

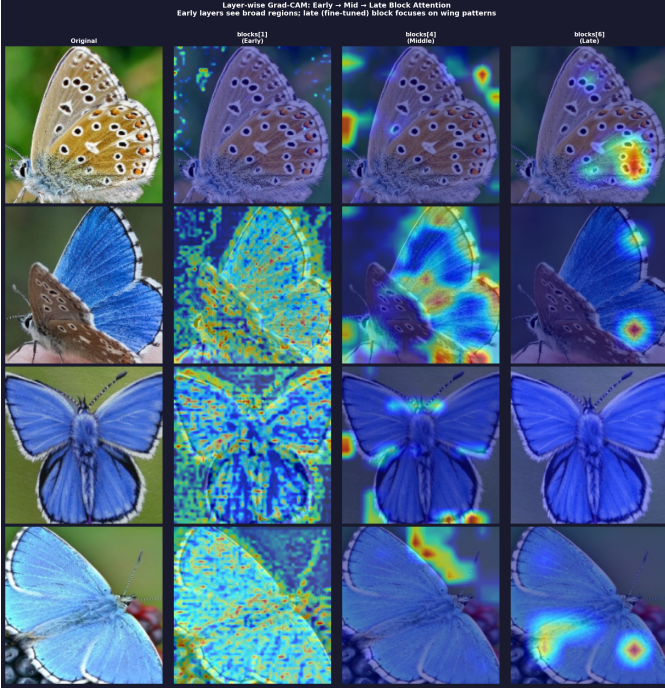


Fig. 4. Layer-wise Grad-CAM at blocks[1] (early), blocks[4] (middle), and blocks[6] (late/fine-tuned) for the same input images. Heatmaps progress from broad whole-image activations to tight, species-specific wing-feature highlights.

blocks[6] (Late — fine-tuned): Activation is tightly concentrated on fine-grained discriminative features — specific wing bands, colour patches, eyespots, and margin patterns. This is precisely the block we unlocked during Phase 2 fine-tuning; its representations have been adapted from ImageNet semantics to butterfly-specific diagnostic anatomy.

Why this matters: The layer-wise progression provides strong empirical evidence that fine-tuning only blocks[6] is sufficient. Blocks 0–5 already encode a rich feature pyramid that transfers well to the butterfly domain. By unlocking only the final block, we “sharpen the lens” for species-level discrimination without destabilising the foundational visual vocabulary learned from ImageNet. This is why Strategy B (2.9 M trainable params) matches Strategy C (5.3 M trainable params) in final test accuracy.

C. UMAP Feature Space

UMAP [5] reduces the 1,280-dimensional penultimate-layer features to 2D while preserving local neighbourhood structure. Figure 5 plots the 500 test images in this reduced space.

The UMAP shows **compact, non-overlapping islands per species**. Biologically related groups (large swallowtails, blue *Lycenidae*) appear as neighbouring macro-clusters, indicating the model has learned a latent space that mirrors real phylogenetic relationships. Contrast this with the CLIP embedding, which showed amorphous, heavily overlapping blobs — directly explaining the 30.8% vs 97.0% accuracy gap.

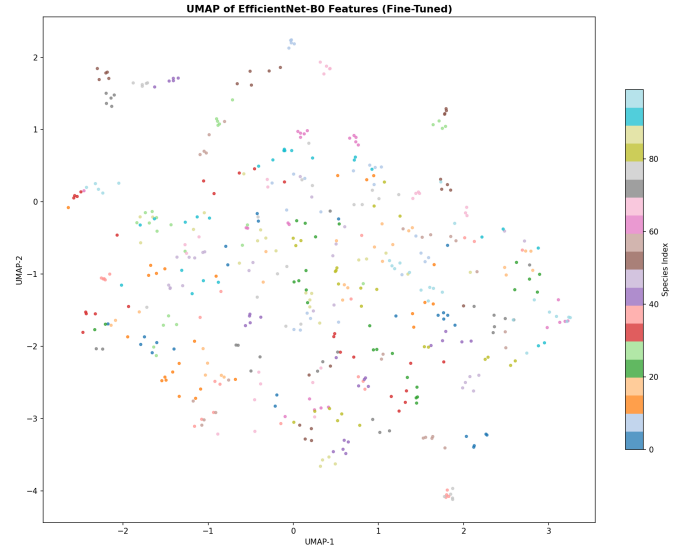


Fig. 5. UMAP of fine-tuned EfficientNet features (500 test images, 100 species). Species form tight, well-separated clusters; biologically similar species (e.g., co-mimics) neighbour each other, reflecting the learned taxonomy.

VI. CONFUSION ANALYSIS

The final test metrics for Strategy B are:

- **Top-1 Accuracy:** 97.0%
- **Top-3 Accuracy:** 99.2%
- **Macro Precision:** 97.36%
- **Macro Recall:** 97.0%
- **Macro F1-score:** 96.92%

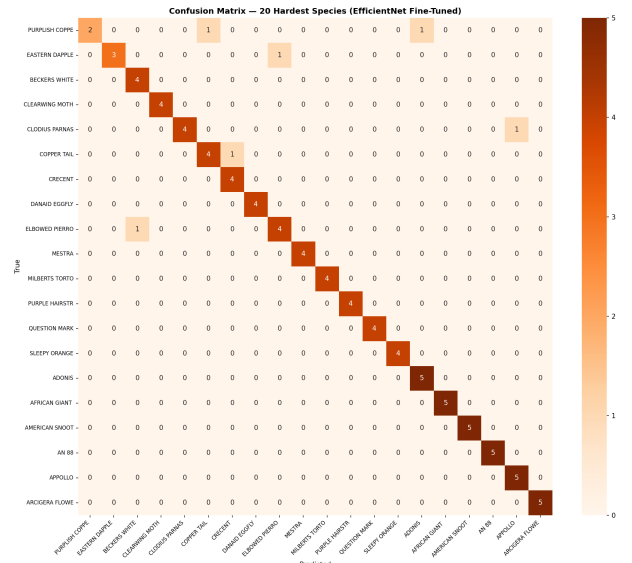


Fig. 6. Confusion matrix restricted to the 20 most error-prone species. Confusions cluster around known mimicry pairs (e.g., Monarch/Viceroy, Adonis/Chalk Hill Blue), not across unrelated taxa.

The confusion matrix (Figure 6) restricts to the 20 highest-error species. All confusions occur between species with

documented mimicry relationships or extreme morphological similarity. There are no cross-family errors (e.g., a Swallowtail mistaken for a Lycaenid), confirming that residual errors are irreducible without higher-resolution imaging.

VII. DEPLOYMENT

The trained weights for all three strategies are served from a public Hugging Face Hub repository (Anurag29104/butterfly-classifier). The Streamlit application downloads the selected model on first request and caches it with `@st.cache_resource`. The sidebar exposes a radio selector to switch between Strategy A, B, and C in real time, with live Top-1, Top-3, and Macro-F1 metrics. All Wikipedia summaries, IUCN conservation statuses, and species fun facts are pre-cached locally to eliminate runtime HTTP latency.

VIII. CONCLUSION

This project demonstrates that for fine-grained scientific classification, zero-shot foundation models are insufficient: CLIP’s 30.8% accuracy reflects its lack of species-level visual vocabulary. By contrast, unlocking only the final convolutional block of EfficientNet-B0 — just 2.9 M of 4.14 M total parameters — achieves **97.0% Top-1 accuracy**, matching a fully fine-tuned model at lower cost. Grad-CAM confirms that the model attends to biologically meaningful wing features, and the UMAP embedding reveals a taxonomically coherent latent space. The complete system is deployed as a real-time Streamlit application.

REFERENCES

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- [7] R. Wightman, “PyTorch Image Models (timm),” GitHub, 2019. [Online]. Available: <https://github.com/huggingface/pytorch-image-models>

APPENDIX

- **Live Project (Streamlit Deployed)** : <https://srushtipekamwar-smai-assignment-3-app-dqqu7l.streamlit.app/>
- **GitHub Repository**: https://github.com/SrushtiPekamwar/SMAI_Assignment_3

- **Hugging Face Model Hub**: <https://huggingface.co/Anurag29104/butterfly-classifier>
- **CLIP Zero-Shot model**: <https://huggingface.co/openai/clip-vit-base-patch32>
- **EfficientNet-B0 via timm**: <https://github.com/huggingface/pytorch-image-models>

Caching strategy: Wikipedia summaries, IUCN conservation status mappings, and biological fun facts are pre-fetched and stored in `data/species_metadata.json` so that the Streamlit app serves rich species information without runtime HTTP requests.

The following screenshots demonstrate the user interface and functionality of the deployed Streamlit application.

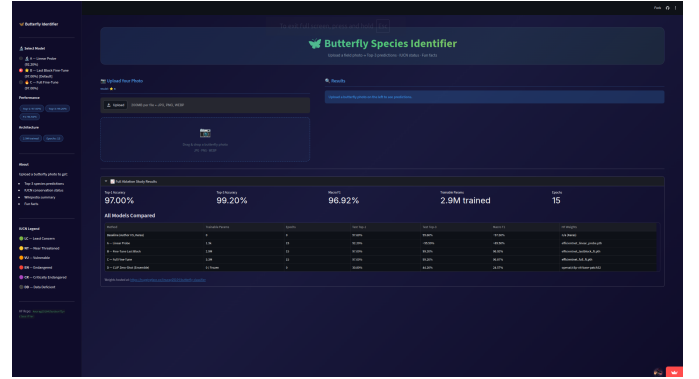


Fig. 7. Landing page of the application, ready for user image upload.

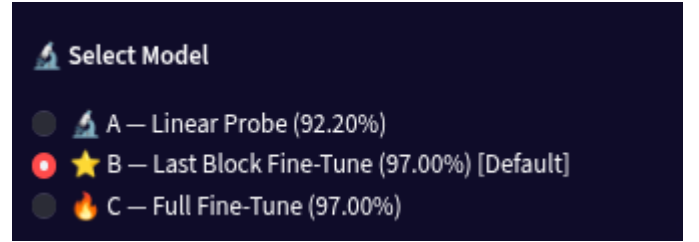


Fig. 8. Model selection dropdown in the sidebar, allowing the user to choose between the Phase 1, Phase 2, and Full Fine-Tuned models.

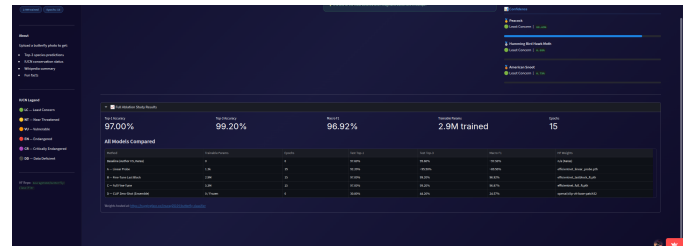


Fig. 9. Top-3 probability results table displayed immediately after inference.

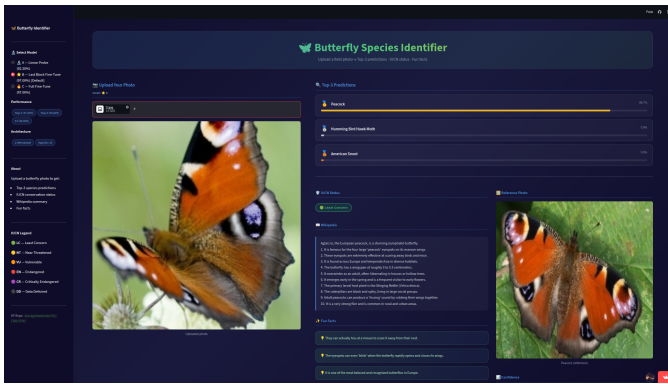


Fig. 10. Dynamic layout showing the model's prediction along with the fetched Wikipedia summary, IUCN status and 3 fun facts.